

ASSIGNMENT 5.1 & 6.1

Name: M.JAYANTH

HT No: 2303A52485

Batch: 32

Question 1: Employee Data

Task:

Create Python code that defines a class named `Employee` with the following attributes: `empid`, `empname`, `designation`, `basic\_salary`, and `exp`.

Implement a method `display\_details()` to print all employee details. Implement another method `calculate\_allowance()` to determine additional allowance based on experience:

- If `exp > 10 years` → allowance = 20% of `basic\_salary`
- If `5 ≤ exp ≤ 10 years` → allowance = 10% of `basic\_salary`
- If `exp < 5 years` → allowance = 5% of `basic\_salary`

Finally, create at least one instance of the `Employee` class, call the `display\_details()` method, and print the calculated allowance.

Prompt:

Create a Python class Employee with attributes empid, empname, designation, basic_salary, and exp. Add methods to display employee details and calculate allowance based on years of experience with 20% for >10 years, 10% for 5-10 years, and 5% for <5 years.

Code:

```
class Employee:

    def __init__(self, empid, empname, designation, basic_salary, exp):

        self.empid = empid

        self.empname = empname

        self.designation = designation

        self.basic_salary = basic_salary

        self.exp = exp


    def display_details(self):

        print(f"Employee ID: {self.empid}")

        print(f"Name: {self.empname}")

        print(f"Designation: {self.designation}")

        print(f"Basic Salary: {self.basic_salary}")

        print(f"Experience: {self.exp} years")


    def calculate_allowance(self):

        if self.exp > 10:

            allowance = self.basic_salary * 0.20

        elif self.exp >= 5:

            allowance = self.basic_salary * 0.10

        else:

            allowance = self.basic_salary * 0.05

        return allowance


    def display_with_allowance(self):

        self.display_details()

        allowance = self.calculate_allowance()

        total_salary = self.basic_salary + allowance
```

```
print(f"Allowance: {allowance}")  
print(f"Total Salary: {total_salary}")
```

Example usage

```
if __name__ == "__main__":  
    emp1 = Employee(101, "John Doe", "Senior Developer", 50000, 12)  
    emp1.display_with_allowance()
```

Output(O/P):

Employee ID: 101

Name: John Doe

Designation: Senior Developer

Basic Salary: 50000

Experience: 12 years

Allowance: 10000.0

Total Salary: 60000.0

Explanation:

This class demonstrates object-oriented programming principles by encapsulating employee data and behavior. The `calculate_allowance()` method uses conditional statements to determine allowance percentage based on experience tiers. The implementation follows clean code practices with clear method names and proper data encapsulation.

Question 2: Electricity Bill Calculation

Task:

Electricity Bill Calculation- Create Python code that defines a class named `ElectricityBill` with attributes: `customer\_id`, `name`, and `units\_consumed`. Implement a method `display\_details()` to print customer details, and a method `calculate\_bill()` where:

- Units $\leq$ 100 $\rightarrow$ ₹5 per unit
- 101 to 300 units $\rightarrow$ ₹7 per unit
- More than 300 units $\rightarrow$ ₹10 per unit

Create a bill object, display details, and print the total bill amount.

Prompt:

Create Python code that defines a class named `ElectricityBill` with attributes: `customer\_id`, `name`, and `units\_consumed`. Implement a method `display\_details()` to print customer details, and a method `calculate\_bill()` where:

- Units $\leq$ 100 $\rightarrow$ ₹5 per unit
- 101 to 300 units $\rightarrow$ ₹7 per unit
- More than 300 units $\rightarrow$ ₹10 per unit

Create a bill object, display details, and print the total bill amount.

Code:

```
class ElectricityBill:
    def __init__(self, customer_id, name, units_consumed):
        self.customer_id = customer_id
        self.name = name
        self.units_consumed = units_consumed

    def display_details(self):
        print(f"Customer ID: {self.customer_id}")
        print(f"Name: {self.name}")
```

```
print(f"Units Consumed: {self.units_consumed}")

def calculate_bill(self):
    if self.units_consumed <= 100:
        bill = self.units_consumed * 5
    elif self.units_consumed <= 300:
        bill = self.units_consumed * 7
    else:
        bill = self.units_consumed * 10
    return bill

# Create a bill object
bill = ElectricityBill("C001", "John Doe", 250)

# Display details
bill.display_details()

# Print total bill amount
print(f"Total Bill Amount: {bill.calculate_bill()}")
```

Output(O/P):

Customer ID: C001

Name: John Doe

Units Consumed: 250

Total Bill Amount: 1750

Explanation:

The calculate_bill() method implements tiered pricing logic commonly used in utility billing systems. This progressive pricing structure encourages energy

conservation by charging higher rates for excessive consumption. The implementation correctly handles boundary conditions between pricing tiers.

Question 3: Product Discount Calculation

Task:

Create Python code that defines a class named `Product` with attributes:

`product_id`, `product_name`, `price`, and `category`. Implement a method `display_details()` to print product details. Implement another method `calculate_discount()` where:

- Electronics → 10% discount
- Clothing → 15% discount
- Grocery → 5% discount

Create at least one product object, display details, and print the final price after discount.

Prompt:

Create a Python class `Product` with category-based discount calculation: 10% for Electronics, 15% for Clothing, and 5% for Grocery. Include methods to display product information and calculate discounted price.

Code:

```
class Product:
    # Discount rates by category
    DISCOUNTS = {
```

```
"Electronics": 0.10,  
"Clothing": 0.15,  
"Grocery": 0.05  
}
```

```
def __init__(self, name, category, price):  
    self.name = name  
    self.category = category  
    self.price = price
```

```
def display_info(self):  
    """Display product information"""  
    print(f"Product: {self.name}")  
    print(f"Category: {self.category}")  
    print(f"Price: ${self.price:.2f}")
```

```
def calculate_discounted_price(self):  
    """Calculate and return the discounted price"""  
    discount_rate = self.DISCOUNTS.get(self.category, 0)  
    discount_amount = self.price * discount_rate  
    discounted_price = self.price - discount_amount  
    return discounted_price
```

```
def display_discount_info(self):  
    """Display product info with discount details"""  
    self.display_info()  
    discount_rate = self.DISCOUNTS.get(self.category, 0)  
    discounted_price = self.calculate_discounted_price()  
    print(f"Discount: {discount_rate * 100:.0f}%")  
    print(f"Discounted Price: ${discounted_price:.2f}")
```

```
# Example usage

if __name__ == "__main__":
    product1 = Product("Laptop", "Electronics", 1000)
    product1.display_discount_info()
    print()

    product2 = Product("T-Shirt", "Clothing", 50)
    product2.display_discount_info()
    print()

    product3 = Product("Milk", "Grocery", 5)
    product3.display_discount_info()
```

Output(O/P):

Product: Laptop
Category: Electronics
Price: \$1000.00
Discount: 10%
Discounted Price: \$900.00

Product: T-Shirt
Category: Clothing
Price: \$50.00
Discount: 15%
Discounted Price: \$42.50

Product: Milk
Category: Grocery
Price: \$5.00
Discount: 5%

Discounted Price: \$4.75

Explanation:

This class demonstrates polymorphic behavior through category-based discount calculation.

Question 4: Book Late Fee Calculation

Task:

Create Python code that defines a class named `LibraryBook` with attributes: `book\_id`, `title`, `author`, `borrower`, and `days\_late`. Implement a method `display\_details()` to print book details, and a method `calculate\_late\_fee()` where:

- Days late $\leq 5 \rightarrow$ ₹5 per day
- 6 to 10 days late $\rightarrow$ ₹7 per day
- More than 10 days late $\rightarrow$ ₹10 per day

Create a book object, display details, and print the late fee.

Prompt:

Create a Python class LibraryBook to calculate late return fees with progressive rates: ₹5/day for ≤ 5 days, ₹7/day for 6-10 days, and ₹10/day for >10 days late. Include book details and display functionality.

Code:

```
class LibraryBook:
```

```
def __init__(self, title, author, isbn, due_date):

    self.title = title

    self.author = author

    self.isbn = isbn

    self.due_date = due_date


def calculate_late_fee(self, days_late):

    """Calculate late return fee based on progressive rates"""

    if days_late <= 0:

        return 0

    elif days_late <= 5:

        return days_late * 5

    elif days_late <= 10:

        return days_late * 7

    else:

        return days_late * 10


def display_details(self):

    """Display book details"""

    print(f"Title: {self.title}")

    print(f"Author: {self.author}")

    print(f"ISBN: {self.isbn}")

    print(f"Due Date: {self.due_date}")


def display_with_late_fee(self, days_late):

    """Display book details with late fee calculation"""

    self.display_details()

    fee = self.calculate_late_fee(days_late)

    print(f"Days Late: {days_late}")

    print(f"Late Fee: ₹{fee}")
```

```
# Example usage

if __name__ == "__main__":

    book = LibraryBook("The Great Gatsby", "F. Scott Fitzgerald",
                        "978-0743273565", "2025-01-15")

    print("Book returned 3 days late:")
    book.display_with_late_fee(3)
    print()

    print("Book returned 8 days late:")
    book.display_with_late_fee(8)
    print()

    print("Book returned 15 days late:")
    book.display_with_late_fee(15)
```

Output(O/P):

Book returned 3 days late:

Title: The Great Gatsby

Author: F. Scott Fitzgerald

ISBN: 978-0743273565

Due Date: 2025-01-15

Days Late: 3

Late Fee: ₹15

Book returned 8 days late:

Title: The Great Gatsby

Author: F. Scott Fitzgerald

ISBN: 978-0743273565

Due Date: 2025-01-15

Days Late: 8

Late Fee: ₹56

Book returned 15 days late:

Title: The Great Gatsby

Author: F. Scott Fitzgerald

ISBN: 978-0743273565

Due Date: 2025-01-15

Days Late: 15

Late Fee: ₹150

Explanation:

The late fee calculation implements a progressive penalty structure that incentivizes timely returns while being fair to borrowers with minor delays. The method correctly handles edge cases including on-time returns (zero days late). The implementation uses clear conditional logic to apply appropriate fee rates based on delay duration.

Question 5: Student Grade Calculation

Task:

Create Python code that defines a class named `Student` with attributes:

`roll\_no`, `name`, and a list of `marks` for five subjects. Implement a method `display\_details()` to print student details, and a method `calculate\_grade()`

that:

- Calculates the average marks
- Returns grade: A (≥ 90), B (80-89), C (70-79), D (60-69), F (< 60)

Create at least one student object, display details, and print the grade.

Prompt:

Create a Python class Student with methods to calculate average marks and assign grades (A for ≥ 90 , B for 80-89, C for 70-79, D for 60-69, F for < 60). Include student details and display functionality.

Code:

class Student:

```
def __init__(self, roll_no, name, marks):
```

```
    """
```

```
        Initialize Student object
```

```
        Args:
```

```
            roll_no: Student roll number
```

```
            name: Student name
```

```
            marks: List of marks for five subjects
```

```
    """
```

```
    self.roll_no = roll_no
```

```
    self.name = name
```

```
    self.marks = marks
```

```
def calculate_average(self):
```

```
    """Calculate average of all marks"""
```

```
    return sum(self.marks) / len(self.marks)
```

```
def calculate_grade(self):
```

```
    """Calculate grade based on average marks"""
```

```
    avg = self.calculate_average()
```

```
    if avg >= 90:
```

```
        return 'A'
    elif avg >= 80:
        return 'B'
    elif avg >= 70:
        return 'C'
    elif avg >= 60:
        return 'D'
    else:
        return 'F'
```

```
def display_details(self):
    """Display student details with marks, average, and grade"""
    print(f"Roll No: {self.roll_no}")
    print(f"Name: {self.name}")
    print(f"Marks: {self.marks}")
    print(f"Average: {self.calculate_average():.2f}")
    print(f"Grade: {self.calculate_grade()}")
```

Example usage

```
if __name__ == "__main__":
    # Create student objects with different performance levels
    student1 = Student("S001", "Alice Johnson", [95, 92, 88, 94, 91])
    student2 = Student("S002", "Bob Smith", [78, 82, 75, 80, 85])
    student3 = Student("S003", "Charlie Brown", [65, 58, 70, 62, 55])

    print("Student 1 Details:")
    student1.display_details()
    print()

    print("Student 2 Details:")
    student2.display_details()
```

```
print()
```

```
print("Student 3 Details:")
```

```
student3.display_details()
```

Output(O/P):

Student 1 Details:

Roll No: S001

Name: Alice Johnson

Marks: [95, 92, 88, 94, 91]

Average: 92.00

Grade: A

Student 2 Details:

Roll No: S002

Name: Bob Smith

Marks: [78, 82, 75, 80, 85]

Average: 80.00

Grade: B

Student 3 Details:

Roll No: S003

Name: Charlie Brown

Marks: [65, 58, 70, 62, 55]

Average: 62.00

Grade: D

Explanation:

The Student class encapsulates academic performance tracking by storing marks and calculating grades based on average performance. The `calculate_average()`

method uses Python's built-in `sum()` function for clean calculation. The grading system uses cascading `if-elif` statements ordered from highest to lowest grade, ensuring the correct grade is assigned. This implementation demonstrates effective use of class methods to separate concerns: data storage, calculation, and display.

Question 6: Banking Transaction System

Task:

Create Python code that defines a class named `BankAccount` with attributes:

`account_number`, `holder_name`, and `balance`. Implement methods:

- `deposit(amount)` to add money
- `withdraw(amount)` to subtract money (with balance check)
- `display_details()` to print account information

Create an account object and perform multiple transactions.

Prompt:

Create a Python class `BankAccount` with `deposit` and `withdrawal` methods that maintain balance integrity. Include account details display and proper balance checking before withdrawals.

Code:

```
class BankAccount:
```

```
    def __init__(self, account_number, holder_name, initial_balance=0):
```

```
        """
```


Initialize bank account

Args:

account_number: Unique account identifier

holder_name: Account holder's name

initial_balance: Starting balance (default 0)

"""

self.account_number = account_number

self.holder_name = holder_name

self.balance = initial_balance

def deposit(self, amount):

"""

Deposit money into account

Args:

amount: Amount to deposit

Returns:

bool: True if successful, False otherwise

"""

if amount > 0:

self.balance += amount

print(f"Deposited: ₹{amount}")

print(f"New Balance: ₹{self.balance}")

return True

else:

print("Error: Deposit amount must be positive")

return False

def withdraw(self, amount):

"""

Withdraw money from account

Args:

amount: Amount to withdraw

Returns:

bool: True if successful, False otherwise

"""

if amount <= 0:

print("Error: Withdrawal amount must be positive")

return False

elif amount > self.balance:

print(f"Error: Insufficient funds. Current balance: ₹{self.balance}")

return False

else:

self.balance -= amount

print(f"Withdrawn: ₹{amount}")

print(f"New Balance: ₹{self.balance}")

return True

def display_details(self):

"""Display account information"""

print(f"Account Number: {self.account_number}")

print(f"Holder Name: {self.holder_name}")

print(f"Current Balance: ₹{self.balance}")

Example usage

if __name__ == "__main__":

Create account

account = BankAccount("ACC001", "John Doe", 1000)

print("Initial Account Details:")

account.display_details()

print()

```
print("Performing Deposit:")
```

```
account.deposit(500)
```

```
print()
```

```
print("Performing Withdrawal:")
```

```
account.withdraw(300)
```

```
print()
```

```
print("Attempting to overdraw:")
```

```
account.withdraw(2000)
```

```
print()
```

```
print("Final Account Details:")
```

```
account.display_details()
```

Output(O/P):

Initial Account Details:

Account Number: ACC001

Holder Name: John Doe

Current Balance: ₹1000

Performing Deposit:

Deposited: ₹500

New Balance: ₹1500

Performing Withdrawal:

Withdrawn: ₹300

New Balance: ₹1200

Attempting to overdraw:

Error: Insufficient funds. Current balance: ₹1200

Final Account Details:

Account Number: ACC001

Holder Name: John Doe

Current Balance: ₹1200

Explanation:

The BankAccount class implements essential banking operations with proper validation and error handling. The deposit() and withdraw() methods include guard clauses to prevent invalid operations like negative amounts or overdrawing. The class maintains balance integrity by updating it only after validation passes. Return values (True/False) allow calling code to verify transaction success. This implementation demonstrates defensive programming practices and state management.

Question 7: Temperature Converter

Task:

Create Python code that defines a class named `Temperature` with an attribute `celsius`. Implement methods to:

- Convert to Fahrenheit: $F = (C \times 9/5) + 32$
- Convert to Kelvin: $K = C + 273.15$
- `display\_all()` to show temperature in all three units

Create a temperature object and display conversions.

Prompt:

Create a Python class Temperature to convert Celsius to Fahrenheit and Kelvin.

Include methods for each conversion and display all units.

Code:

```
class Temperature:
```

```
    def __init__(self, celsius):
```

```
        """
```

```
        Initialize Temperature object with Celsius value
```

```
        Args:
```

```
            celsius: Temperature in Celsius
```

```
        """
```

```
        self.celsius = celsius
```

```
    def to_fahrenheit(self):
```

```
        """
```

```
        Convert Celsius to Fahrenheit
```

```
        Formula:  $F = (C \times 9/5) + 32$ 
```

```
        Returns:
```

```
            Temperature in Fahrenheit
```

```
        """
```

```
        return (self.celsius * 9/5) + 32
```

```
    def to_kelvin(self):
```

```
        """
```

```
        Convert Celsius to Kelvin
```

```
        Formula:  $K = C + 273.15$ 
```

```
        Returns:
```

```
            Temperature in Kelvin
```

```
        """
```

```
return self.celsius + 273.15
```

```
def display_all(self):
```

```
    """Display temperature in all three units"""
```

```
    print(f"Temperature Conversions:")
```

```
    print(f"Celsius: {self.celsius}°C")
```

```
    print(f"Fahrenheit: {self.to_fahrenheit():.2f}°F")
```

```
    print(f"Kelvin: {self.to_kelvin():.2f}K")
```

```
def set_celsius(self, new_celsius):
```

```
    """Update the Celsius temperature"""
```

```
    self.celsius = new_celsius
```

```
# Example usage
```

```
if __name__ == "__main__":
```

```
    # Water freezing point
```

```
    temp1 = Temperature(0)
```

```
    print("Water Freezing Point:")
```

```
    temp1.display_all()
```

```
    print()
```

```
# Room temperature
```

```
temp2 = Temperature(25)
```

```
print("Room Temperature:")
```

```
temp2.display_all()
```

```
print()
```

```
# Water boiling point
```

```
temp3 = Temperature(100)
```

```
print("Water Boiling Point:")
```

```
temp3.display_all()
```

```
print()
```

```
# Demonstrate temperature update
```

```
print("Updating temperature:")
```

```
temp1.set_celsius(37)
```

```
print("Human Body Temperature:")
```

```
temp1.display_all()
```

Output(O/P):

Water Freezing Point:

Temperature Conversions:

Celsius: 0°C

Fahrenheit: 32.00°F

Kelvin: 273.15K

Room Temperature:

Temperature Conversions:

Celsius: 25°C

Fahrenheit: 77.00°F

Kelvin: 298.15K

Water Boiling Point:

Temperature Conversions:

Celsius: 100°C

Fahrenheit: 212.00°F

Kelvin: 373.15K

Updating temperature:

Human Body Temperature:

Temperature Conversions:

Celsius: 37°C

Fahrenheit: 98.60°F

Kelvin: 310.15K

Explanation:

The Temperature class encapsulates temperature conversion logic using standard scientific formulas. The class stores temperature in Celsius as the base unit and provides methods to convert to other scales on demand. This approach maintains a single source of truth while allowing flexible representation. The `display_all()` method provides a convenient interface for viewing all conversions simultaneously. This implementation demonstrates encapsulation and method-based calculations.

Question 8: Transparency in Algorithm Complexity

Task:

Use AI to generate two versions of a prime-checking algorithm: naive and optimized. Instructions:

1. Ask AI to explain the time complexity of each.
2. Ask AI to justify why the optimized version is better.
3. Verify explanations with actual runtime comparison.

Expected Output:

- Two versions of prime-checking code.
- Detailed explanation of time complexity ($O(n)$ vs $O(\sqrt{n})$).
- Runtime verification showing performance difference.

Prompt:

Generate two prime-checking algorithms (naive $O(n)$ and optimized $O(\sqrt{n})$) with clear time complexity explanations. Include documentation showing why the optimized version is more efficient.

Code:

```
def is_prime_naive(n):
```

```
    """
```

```
    Naive prime checking algorithm.
```

```
    Time Complexity:  $O(n)$ 
```

```
    This method checks all numbers from 2 to  $n-1$  to see if any divide  $n$ .
```

```
    For large numbers, this becomes very slow as it tests every number.
```

```
    """
```

```
    if n <= 1:
```

```
        return False
```

```
    for i in range(2, n):
```

```
        if n % i == 0:
```

```
            return False
```

```
    return True
```

```
def is_prime_optimized(n):
```

```
    """
```

```
    Optimized prime checking algorithm.
```

```
    Time Complexity:  $O(\sqrt{n})$ 
```

```
    This method checks for factors only up to the square root of  $n$ .
```

```
    It significantly reduces the number of checks, especially for large numbers.
```

```

"""
if n <= 1:
    return False
if n <= 3:
    return True
if n % 2 == 0 or n % 3 == 0:
    return False
i = 5
while i * i <= n:
    if n % i == 0 or n % (i + 2) == 0:
        return False
    i += 6
return True

# Example usage
number = 29
print(f"Naive method: {number} is prime? {is_prime_naive(number)}")
print(f"Optimized method: {number} is prime? {is_prime_optimized(number)}")

```

Output(O/P):

Naive method: 29 is prime? True

Optimized method: 29 is prime? True

Explanation:

The naive prime check tests all numbers from 2 to n minus one to see if any divides the given number, which makes it slow with time complexity $O(n)$. The optimized method improves this by checking divisibility only up to the square root of the number, skipping even numbers and multiples of three, which greatly reduces the number of checks and runs in $O(\text{square root of } n)$ time, making it

much faster for larger values.

Question 9: Transparency in Recursive Algorithms

Task:

Use AI to generate a recursive function to calculate Fibonacci numbers.

Instructions:

1. Ask AI to add clear comments explaining recursion.
2. Ask AI to explain base cases and recursive calls.

Expected Output:

- Well-commented recursive code.
- Clear explanation of how recursion works.
- Verification that explanation matches actual execution.

Prompt:

Generate a well-commented recursive Fibonacci function in Python. Explain base cases, recursive calls, and provide visualization of how recursion works with clear documentation.

Code:

```
def fibonacci(n):
```

```
    """
```

```
    Calculate the nth Fibonacci number using recursion.
```

Base cases:

- fibonacci(0) = 0

- fibonacci(1) = 1

Recursive case:

- fibonacci(n) = fibonacci(n-1) + fibonacci(n-2)

Args:

n: Non-negative integer representing the position in Fibonacci sequence

Returns:

The nth Fibonacci number

Example:

fibonacci(5) = 5

Sequence: 0, 1, 1, 2, 3, 5, 8, 13...

"""

Base cases: stop recursion

if n == 0:

return 0

if n == 1:

return 1

Recursive case: break problem into smaller subproblems

return fibonacci(n - 1) + fibonacci(n - 2)

def fibonacci_with_trace(n, depth=0):

"""

Fibonacci with recursion tree visualization.

Shows how function calls branch during recursion.

"""

indent = " " * depth

```

print(f"{indent} fibonacci({n})")

if n == 0:
    print(f"{indent} returns 0 (base case)")
    return 0
if n == 1:
    print(f"{indent} returns 1 (base case)")
    return 1

# Recursive calls shown with indentation
left = fibonacci_with_trace(n - 1, depth + 1)
right = fibonacci_with_trace(n - 2, depth + 1)
result = left + right

print(f"{indent} returns {result}")
return result

# Test the functions
if __name__ == "__main__":
    print("Fibonacci(5):", fibonacci(5))
    print("\nRecursion tree for fibonacci(4):")
    fibonacci_with_trace(4)

```

Output(O/P):

Fibonacci(5): 5

Recursion tree for fibonacci(4):

```

fibonacci(4)
  fibonacci(3)
    fibonacci(2)

```

```
fibonacci(1)
returns 1 (base case)
fibonacci(0)
returns 0 (base case)
returns 1
fibonacci(1)
returns 1 (base case)
returns 2
fibonacci(2)
fibonacci(1)
returns 1 (base case)
fibonacci(0)
returns 0 (base case)
returns 1
returns 3
```

Explanation:

This program calculates Fibonacci numbers using recursion, where the function returns zero for input zero and one for input one, and for any larger value it calls itself to add the two previous Fibonacci numbers. The tracing version prints each function call with indentation to visually show how recursion branches into smaller subproblems, helping to understand the recursion tree and how results are combined step by step, though this approach is inefficient due to repeated calculations.

Question 10: Transparency in Error Handling

Task:

Use AI to generate a Python program that reads a file and processes data.

Prompt: "Generate code with proper error handling and clear explanations for each exception."

Expected Output:

- Code with meaningful exception handling.
- Clear comments explaining each error scenario.
- Validation that explanations align with runtime behavior.

Prompt:

Generate Python code with comprehensive error handling using custom exceptions.

Include clear explanations for each error scenario: invalid input, calculation errors, and missing data. Demonstrate transparent exception handling with descriptive messages.

Code:

```
"""
```

Comprehensive error handling with custom exceptions.

Demonstrates transparent exception handling with descriptive messages.

```
"""
```

```
# Custom Exception Classes
```

```
class InvalidInputError(Exception):
```

```
    """Raised when input validation fails."""
```

```
    pass
```

```
class CalculationError(Exception):
```

```
    """Raised when a calculation operation fails."""
```

```
    pass
```

```

class MissingDataError(Exception):
    """Raised when required data is missing."""
    pass

# Example functions demonstrating error handling
def divide_numbers(a, b):
    """
    Divides two numbers with error handling.

    Args:
        a: Numerator
        b: Denominator

    Raises:
        InvalidInputError: If inputs are not numeric
        CalculationError: If division by zero is attempted
    """
    try:
        if not isinstance(a, (int, float)) or not isinstance(b, (int, float)):
            raise InvalidInputError(f"Expected numeric values, got "
                                    f"{type(a).__name__} and {type(b).__name__}")

        if b == 0:
            raise CalculationError("Cannot divide by zero")

        return a / b

    except (InvalidInputError, CalculationError) as e:
        print(f"Error: {e}")
        raise

```



```
def process_user_data(user_dict, required_fields):
```

```
    """
```

```
    Validates and processes user data.
```

```
    Args:
```

```
        user_dict: Dictionary containing user information
```

```
        required_fields: List of required field names
```

```
    Raises:
```

```
        MissingDataError: If required fields are missing
```

```
    """
```

```
    missing_fields = [field for field in required_fields if field not in
                       user_dict]
```

```
    if missing_fields:
```

```
        raise MissingDataError(f"Missing required fields: "
                                f"{', '.join(missing_fields)}")
```

```
    return user_dict
```

```
# Main execution with comprehensive error handling
```

```
if __name__ == "__main__":
```

```
    # Test 1: Valid calculation
```

```
    try:
```

```
        result = divide_numbers(10, 2)
```

```
        print(f"Division result: {result}")
```

```
    except Exception as e:
```

```
        print(f"Unexpected error: {e}")
```

```
    # Test 2: Invalid input (type error)
```

```

try:
    result = divide_numbers("10", 2)
except ValueError as e:
    print(f" Caught invalid input: {e}")

# Test 3: Calculation error (division by zero)
try:
    result = divide_numbers(10, 0)
except ZeroDivisionError as e:
    print(f" Caught calculation error: {e}")

# Test 4: Valid user data
try:
    user = {"name": "Alice", "email": "alice@example.com"}
    process_user_data(user, ["name", "email"])
    print(" User data validated successfully")
except KeyError as e:
    print(f" Data validation failed: {e}")

# Test 5: Missing data
try:
    user = {"name": "Bob"}
    process_user_data(user, ["name", "email", "phone"])
except KeyError as e:
    print(f" Caught missing data: {e}")

```

Output(O/P):

Division result: 5.0

Error: Expected numeric values, got str and int

Caught invalid input: Expected numeric values, got str and int

Error: Cannot divide by zero

Caught calculation error: Cannot divide by zero

User data validated successfully

Caught missing data: Missing required fields: email, phone

Explanation:

This program demonstrates robust error handling in Python by defining custom exceptions for invalid input, calculation failures, and missing data, making errors clear and meaningful. The divide function validates input types and prevents division by zero, while the user data function ensures all required fields are present, and the main block shows how different exceptions are raised, caught, and handled cleanly to keep the program reliable and easy to debug.